Verified Localized Episodic Grounding for Network Analysis: A Controlled Mechanism Study

Ye Tang[1,†], Ruikai Han[1,†], Yi Hu[1], and Ruzhen Chen[2,*]

[1]School of Information Science and Technology, Xizang University, Lhasa 850000, China
[2]School of Education, Xizang University, Lhasa 850000, China

***Abstract*—Memory-augmented language models leverage historical evidence to support network and diagnostic classification without parameter updates, yet compressing that evidence into reusable verbal explanations often omits boundary exceptions or admits unsupported rules. Existing paradigms emphasize raw episodic retrieval, local reflection with feedback, or global semantic consolidation, motivating controlled comparisons across abstraction units, correction signals, and grounding strategies. We propose Verified Localized Episodic Grounding (VLEG), which validates each candidate reflection against its labeled historical source record prior to memory admission. Within a bounded refinement budget K, separate label-blind generator and label-informed verifier calls to the same backbone revise rejected candidates under a fail-closed Boolean gate and store the first accepted reflection alongside its source episode for hybrid inference. Controlled symbolic diagnostics at 7B and 70B parameter scales demonstrate that iterative verification restores memory-bank coverage and guarantees 100% source-label consistency, enabling VLEG to reach 91.67% accuracy on held-out attribute combinations at 70B while exposing dual-modality attention competition at 7B and single-record blindness to co-occurring distractors. Across five historical-memory seeds on 1,000 shared UNSW-NB15 test flow fingerprints, VLEG improves Macro-F1 by 17.02 percentage points (pp) over unverified first reflections from 42.66% to 59.68% by recovering attack recall from 6.17% to 77.74%, yet trails raw flow retrieval at 63.92% by 4.24 pp within a paired conditional 95% confidence interval of $[-5.91, -2.53]$ pp. Deterministic screening and an admitted-mask raw control reveal that 72.0% of locally valid rules generalize necessary single-flow states into globally sufficient triggers that match opposite-label counterexamples and raise the false positive rate from 38.13% to 48.78%. These findings decouple local source fidelity from cross-episode validity and motivate integrating multi-instance counterexample screening into reflective memory architectures.**

***Index Terms*—Episodic memory, hallucination mitigation, large language models, network traffic analysis, verified memory consolidation**

Supported by the Science and Technology Projects of Ngari Prefecture, Xizang Autonomous Region, China (QYXTZX-AL2026-01).

[1]School of Information Science and Technology, Xizang University, Lhasa 850000, China.

[2]School of Education, Xizang University, Lhasa 850000, China.

Email addresses: Ye Tang: 3441691137@qq.com; Ruikai Han: 1763683932@qq.com; Yi Hu: hy610400@163.com; Ruzhen Chen: crz@xzu.edu.cn.

[†]Ye Tang and Ruikai Han contributed equally to this work and share first authorship.

[*]Corresponding author: Ruzhen Chen (crz@xzu.edu.cn).

I. Introduction

A. Background and Motivation

Large language models can leverage historical examples to support network traffic classification and diagnostic analysis without parameter updates [1], [2]. In operational network security, historical flow records and incident logs provide critical evidence for interpreting emerging traffic anomalies, yet their utility depends on retaining boundary conditions while avoiding unsupported generalizations. Because finite context windows degrade when flooded with unprocessed records, selecting and organizing historical evidence is a central systems challenge [3], [4].

Persistent-memory architectures extend access to historical experience beyond the immediate prompt [5]–[7]. While raw records preserve granular feature values, verbal reflections and summaries offer compact decision guidance at the risk of omitting exceptions or turning incidental covariates into global rules. We investigate this fundamental trade-off by combining controlled diagnostic testbeds with a public network flow benchmark.

The tension between raw retention and verbal compression becomes acute when a model must classify a new observation from a compact support bank of labeled episodes. Retaining raw records preserves exception boundaries but forces the inference model to re-derive discriminative rules on the fly, whereas compressing examples into natural-language rules shifts the reasoning burden to memory construction where unverified abstractions may distort the underlying data. Designing an effective episodic memory therefore requires jointly governing how candidate rules are verified prior to admission and how raw evidence is paired with abstractions during inference.

Crucially, local source agreement and cross-episode validity represent distinct requirements. A candidate rule can accurately describe a single historical record while asserting an overly broad condition that fails across other episodes, such as citing a routine protocol state that merely coincides with an attack label. Because auditing a candidate rule against a single source record cannot detect cross-sample counterexamples, memory systems must be evaluated across admission coverage, source-label fidelity, rule specificity, and downstream classification rather than relying on admission rate alone.

B. Memory Research and Study Positioning

Memory-augmented language models organize experience through episodic retrieval, local reflection, and global semantic consolidation [8], [9], differing in their unit of abstraction, correction signal, and inference grounding. Whereas feedback-driven agents revise actions across sequential trials [10], [11] and global hypothesis frameworks induce dataset-wide rules [12], [13], our study isolates the interaction between per-episode admission verification and hybrid episodic-semantic pairing. Rather than assuming that local verification or hybrid grounding is universally superior, we benchmark VLEG against both pure raw retrieval and pure abstract memory across model scales and domains.

Each memory paradigm exhibits distinct operating characteristics depending on feature dimensionality and model capacity. Episodic retrieval preserves exact feature tuples, local reflection distills per-episode heuristics, and semantic consolidation merges patterns across records, while hybrid grounding exposes both raw episodes and local rules at inference. Understanding when hybrid grounding aids or hinders classification requires decoupling retrieval selection from context formatting and separating iterative critique from simple rejection sampling.

C. Proposed VLEG Method

We formulate Verified Localized Episodic Grounding (VLEG) to audit candidate local reflections before memory admission while retaining their originating episodic evidence. During offline bank construction, a label-blind generator proposes a tentative classification and local rule from unlabeled episode features, and a label-informed verifier audits the candidate against the ground-truth source record. Rejected proposals receive natural-language critique across at most K refinement steps, where the first accepted rule is admitted alongside its raw source episode and persistently rejected episodes are excluded under a fail-closed policy.

Although generator and verifier invocations share backbone weights, strict information-access boundaries withhold historical labels during initial synthesis and withhold test labels entirely during inference. This single-instance admission gate eliminates source-label inversions and unsupported attribute fabrication, even though single-record consistency alone cannot guarantee causal invariance across unseen episodes.

During inference, VLEG retrieves top-k admitted memory tuples to construct a hybrid prompt containing both raw historical records and their verified reflections. In our public network flow evaluation, all local memory conditions share identical label-free feature retrieval indices, ensuring that downstream performance differences reflect memory consolidation and grounding rather than divergent neighbor selection.

D. Experimental Evaluation

To disentangle overlapping failure modes before testing on noisy network telemetry, our evaluation progresses from controlled symbolic diagnostics at 7B and 70B parameter scales to the public UNSW-NB15 network intrusion benchmark. Because the symbolic testbed provides an exact causal oracle with known distractor attributes and held-out attribute combinations, it cleanly separates memory-bank coverage recovery from rule purification and reveals how model scale governs the ability to fuse raw records with abstract rules.

Transferring the same auditing pipeline to deduplicated UNSW-NB15 flow records evaluates whether single-flow verification generalizes to realistic traffic distributions. By combining multi-seed paired bootstrap intervals, an admitted-mask raw retrieval control that strips reflection text while preserving the exact VLEG admission mask, and a deterministic rule compiler that screens admitted IF/THEN rules against historical counterexamples, our experiments explain why single-instance verification recovers attack recall relative to unverified reflection while inflating false alarms relative to raw flow retrieval. We complement these accuracy and latency measurements with a transparent provenance audit covering parser strictness and human review status.

E. Paper Organization

The remainder of this paper is organized as follows. Section II reviews episodic retrieval, local reflection, and semantic consolidation. Section III formalizes the VLEG architecture, detailing its label-isolated verification loop, fail-closed Boolean admission gate, and hybrid inference grounding. Section IV presents controlled symbolic diagnostics, public UNSW-NB15 traffic evaluations, deterministic counterexample screening, and methodological limitations. Finally, Section V summarizes our core mechanistic findings and provides actionable deployment guidelines for network analysis.

II. Related Work

A. Episodic Memory and Retrieval

In-context learning adapts language models through prompt demonstrations without weight updates [1], while retrieval-augmented generation conditions outputs on dynamically retrieved external records [2] and LongMem scales historical access through dedicated memory banks [6]. Complemented by demonstration-selection strategies [14], [15] and context-utilization analyses [3], [8], [9], these non-parametric approaches preserve granular historical evidence without forcing premature rule abstraction.

For structured classification, raw episodic retrieval serves as a natural evidence-preserving baseline because each retrieved exemplar retains its exact feature vector and ground-truth label. While raw exemplars avoid summarization distortion, they require the inference backbone to infer decision boundaries across multiple neighbors within a finite context budget. Our raw-memory baselines follow this principle by supplying labeled historical episodes without offline reflection calls, and our public-flow benchmark locks retrieval indices across local methods to isolate context representation from neighbor selection.

B. Local Reflection and Feedback

Reflective memory architectures extend raw storage by synthesizing verbal guidance from past episodes, as demonstrated

in MemoryBank [16], Generative Agents [17], Reflexion [10], Voyager [11], and structured memory managers [18], [19]. While environmental feedback in sequential agents flags failed rollouts across trials, offline memory verification audits whether a candidate explanation is factually consistent with a static historical record. Neither mechanism automatically guarantees causal sufficiency, because a revised rule can match its source label while still anchoring on an incidental covariate or overly broad numeric threshold.

VLEG formalizes a bounded per-episode admission loop that pairs label-blind initial generation with label-informed verification and critique. By comparing VLEG against both unverified single-step reflection and binary-feedback refinement without natural-language critique, our experiments isolate how explicit verifier feedback shapes memory coverage, source-label fidelity, and downstream generalization.

C. Semantic Consolidation and Hybrid Grounding

Global semantic consolidation compresses an entire experience pool into dataset-wide hypotheses or summary rules [12], [13]. While effective when the underlying domain obeys a compact universal rule, a fixed-length global summary risks erasing minority traffic regimes or boundary exceptions in heterogeneous telemetry. Hybrid memory architectures instead seek to combine abstract rules with concrete episodic exemplars [8], [9], motivating our controlled ablation across raw-only, abstract-only, and hybrid grounding modalities.

III. The Proposed Method

A. Overview of Verified Localized Episodic Grounding (VLEG)

Given a historical support pool of labeled episodes $\mathcal{E} = \{e_1, e_2, \ldots, e_n\}$ where each episode $e_i = (x_i, y_i)$ pairs raw feature attributes x_i with a ground-truth label y_i, pure episodic memory retains $\mathcal{M}_{\text{raw}} = \mathcal{E}$ without abstraction. Conversely, global semantic consolidation [12] distills $\mathcal{E}$ into a compact set of universal hypotheses $\mathcal{H}^*$ maximizing global data likelihood:

$$\mathcal{H}^* = \arg\max_{\mathcal{H}} P(\mathcal{H} \mid e_1, e_2, \ldots, e_n). \tag{1}$$

While a single conditional hypothesis can express simple contextual rules, fixed-budget global summaries often omit local boundary exceptions across complex traffic distributions.

VLEG bridges these extremes by shifting the unit of abstraction from the global population to the individual episode e_i and enforcing a model-based source-episode admission gate. Rather than inducing a monolithic $\mathcal{H}^*$, VLEG populates its memory bank with verified episodic-semantic tuples:

$$\mathcal{M}_{\text{VLEG}} = \{(e_i, r_i^*) \mid e_i \in \mathcal{E},\ \mathcal{V}_\phi(e_i, r_i^*)_v = 1\}, \tag{2}$$

where r_i^* is a localized diagnostic rule derived from episode e_i, and $\mathcal{V}_\phi(e_i, r_i^*)_v \in \{0, 1\}$ denotes the binary acceptance decision of the strict verifier under its checking prompt.

Storing the source episode e_i alongside its validated reflection r_i^* preserves granular feature evidence alongside local guidance, allowing downstream ablations to isolate the effect of each component.

B. Framework Components: Generator, Verifier, and Memory Bank

VLEG comprises three core modules, namely the reflection generator $\mathcal{G}_\theta$, the strict verifier $\mathcal{V}_\phi$, and the hybrid memory bank $\mathcal{M}$. Although generation and verification invoke the same underlying backbone via separate role-specific prompts, strict information-access boundaries separate their inputs.

1) The Reflection Generator ($\mathcal{G}_\theta$): The reflection generator $\mathcal{G}_\theta$ distills a high-dimensional episode $e_i = (x_i, y_i)$ into a localized natural-language rule r_i. To prevent the generator from trivially echoing the known historical label and fabricating a post-hoc rationale, initial synthesis at iteration $j = 0$ is conditioned strictly on the label-free feature attributes x_i:

$$r_i^{(0)} \sim \mathcal{G}_\theta(\cdot \mid x_i). \tag{3}$$

If candidate $r_i^{(j-1)}$ is rejected by the verifier, subsequent refinement iterations ($1 \le j < K$) condition the generator on the raw features x_i, the rejected candidate $r_i^{(j-1)}$, and the diagnostic critique $c_i^{(j-1)}$:

$$r_i^{(j)} \sim \mathcal{G}_\theta(\cdot \mid x_i, r_i^{(j-1)}, c_i^{(j-1)}). \tag{4}$$

Whereas the unverified single-step baseline commits $r_i^{(0)}$ unconditionally and binary-feedback Reflexion [10] revises proposals using scalar correctness signals before committing unconditionally, VLEG admits a reflection only if it passes explicit source verification.

2) The Strict Verifier ($\mathcal{V}_\phi$) and Fail-Closed Boolean Gate: The verifier $\mathcal{V}_\phi$ audits candidate reflection $r_i^{(j)}$ against the complete labeled source episode $e_i = (x_i, y_i)$, checking for label agreement, factual feature support without hallucinated attributes, and testable rule structure:

$$(v_i^{(j)}, c_i^{(j)}) = \mathcal{V}_\phi(e_i, r_i^{(j)}), \tag{5}$$

where $v_i^{(j)} \in \{0, 1\}$ is the binary admission decision and $c_i^{(j)}$ is the natural-language critique returned upon rejection. In our hardened implementation, $\mathcal{V}_\phi$ enforces a fail-closed native-Boolean JSON parser that sets $v_i^{(j)} = 1$ if and only if the verifier output parses as a single JSON object without duplicate keys and its `valid` field is a native Boolean true (type(obj[`"valid"`]) $\equiv$ `bool`), rejecting string literals such as `"false"` or malformed responses.

3) The Hybrid Memory Bank ($\mathcal{M}$): When a candidate passes verification at iteration $j^* \in \{0, \ldots, K-1\}$, the loop terminates immediately and commits the verified reflection $r_i^* = r_i^{(j^*)}$ together with e_i into the hybrid memory bank:

$$\mathcal{M} = \bigcup_{i=1}^{n} \left\{ (e_i, r_i^*) \;\middle|\; \exists\, 0 \le j < K \text{ s.t. } \mathcal{V}_\phi(e_i, r_i^{(j)})_v = 1 \right\}. \tag{6}$$

Episodes that fail verification across all K attempts are omitted from $\mathcal{M}$ under our fail-closed admission policy, defining the admitted index mask $\mathcal{I}_{\text{adm}} \subseteq \{1, \ldots, n\}$.

C. Learning Mechanism: Iterative Cyclic Refinement and Structural Boundary

The iterative cyclic refinement loop couples label-blind generation with label-informed verification across at most K

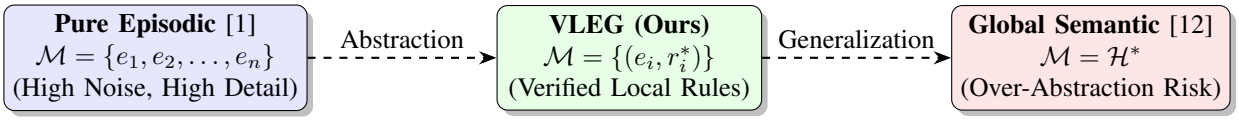


Fig. 1. The theoretical positioning of VLEG within the memory abstraction spectrum. VLEG bridges the gap between pure episodic retention of raw traffic and global semantic consolidation by pairing raw episodes with accepted local abstractions.

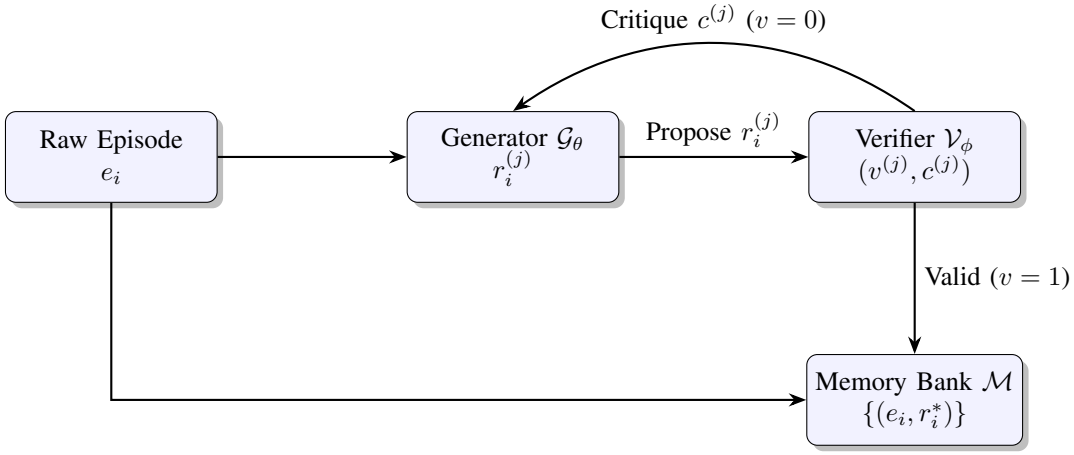


Fig. 2. Architectural flow of the VLEG framework components. Separate generator and verifier calls admit only candidates accepted by the implemented check to the Hybrid Memory Bank alongside their raw episodic counterparts.

attempts per historical episode. Starting from the label-free proposal $r_i^{(0)}$, each rejected candidate ($v_i^{(j)} = 0$) triggers a targeted critique $c_i^{(j)}$ that identifies specific label inversions or unsupported feature claims, guiding the next revision $r_i^{(j+1)}$ until the first accepted reflection is committed or the budget K is exhausted.

Crucially, Equation (5) formalizes the structural boundary of single-instance verification: because $\mathcal{V}_\phi(e_i, r_i^{(j)})$ observes only the originating episode e_i without access to the broader population $\mathcal{E} \setminus \{e_i\}$, it can guarantee local factual consistency on e_i but cannot determine whether a co-occurring attribute or numeric threshold cited in $r_i^{(j)}$ is causally invariant across other episodes or merely an incidental property of e_i.

Remark III.1 (Scope of the admission check). Let A_j denote reaching attempt j, E_j an erroneous candidate at that attempt, and B_j its acceptance by the verifier. Using one-based attempt indices for $j = 1, \ldots, K$, suppose $\Pr(E_j \mid A_j) \le p_h$ and $\Pr(B_j \mid A_j, E_j) \le p_v$ for a specified error class. Because the loop commits the first accepted candidate, the union bound yields

$$\begin{aligned} \Pr(\text{erroneous commitment}) &\le \sum_{j=1}^{K} \Pr(A_j \cap E_j \cap B_j) \\ &\le \min\{1, K p_h p_v\}. \end{aligned} \tag{7}$$

This bound requires no independence assumption across attempts, clarifies that increasing K restores admission coverage rather than monotonically reducing per-admission error, and distinguishes single-pass verification from repeated independent checks on a fixed candidate.

D. Inference Integration and Diagnostic Control Operators

Once the memory bank $\mathcal{M}$ is constructed, VLEG applies *Hybrid Episodic-Semantic Grounding* to classify novel target queries T. Rather than relying solely on unprocessed exemplars [2], [14] or ungrounded rules, hybrid grounding supplies both the raw historical episode e_i and its verified local reflection r_i^* within the inference context [8], [9].

1) Semantic-Driven and Feature-Controlled Retrieval: Given a target query T, VLEG retrieves the top-k relevant tuples from $\mathcal{M}$ using dense embedding cosine similarity via a pre-trained encoder $f_{\text{emb}}(\cdot)$:

$$S(T, \mathcal{M}_i) = \frac{f_{\text{emb}}(T) \cdot f_{\text{emb}}(r_i^*)}{\|f_{\text{emb}}(T)\| \|f_{\text{emb}}(r_i^*)\|}, \tag{8}$$

selecting the subset $\mathcal{M}_{top}$ of size $\min(k, |\mathcal{M}|)$ that maximizes total similarity:

$$\mathcal{M}_{top} = \underset{\substack{\mathcal{S} \subseteq \mathcal{M} \\ |\mathcal{S}| = \min(k, |\mathcal{M}|)}}{\arg\max} \sum_{(e_i, r_i^*) \in \mathcal{S}} S(T, (e_i, r_i^*)). \tag{9}$$

While the symbolic formulation ranks admitted tuples by reflection embeddings $f_{\text{emb}}(r_i^*)$, our public network flow benchmark ranks historical records using label-free feature embeddings $f_{\text{emb}}(x_i)$ across all local methods and removes rejected episodes without replacement, ensuring that neighbor selection is strictly controlled.

2) Hybrid Prompt Formulation and Diagnostic Controls: From the retrieved subset $\mathcal{M}_{top}$, the inference prompt P concatenates each paired episode and verified rule with the target query T:

$$\begin{aligned} Q_i &= \big[\text{Episode: } e_i \oplus \text{Verified Rule: } r_i^*\big], \\ P &= \text{PromptBuilder}\Big(\bigoplus_{(e_i, r_i^*) \in \mathcal{M}_{top}} Q_i \oplus \text{Target: } T \Big), \end{aligned} \tag{10}$$

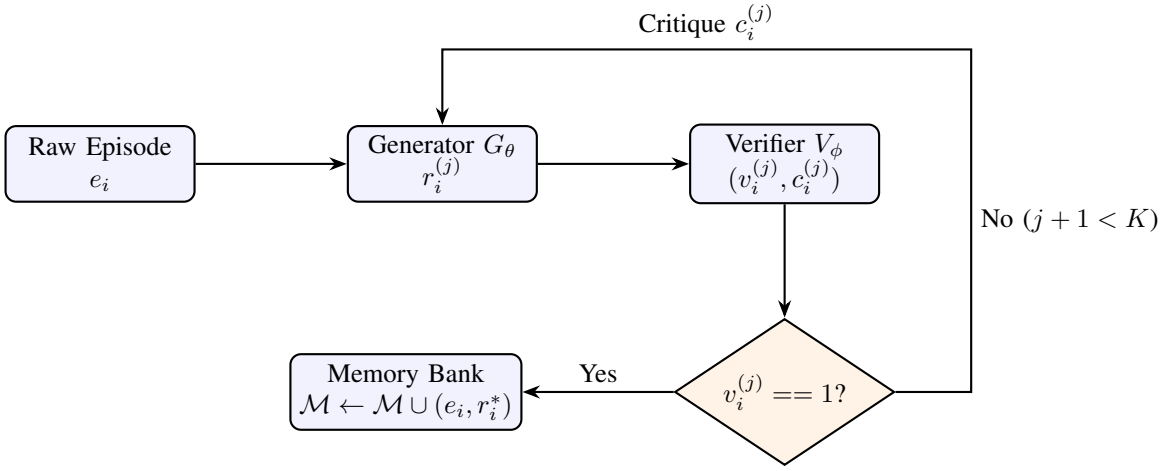


Fig. 3. State transition diagram of the Iterative Cyclic Refinement mechanism. The loop stores the first candidate accepted by the verifier to the paired episodic–semantic memory bank.

where $\oplus$ denotes sequence concatenation, and the reasoning backbone predicts $\hat{y} = \text{LLM}_\psi(P)$. To isolate the mechanisms governing VLEG performance, we pair hybrid inference with two diagnostic operators:

- **Admitted-Mask Raw Control (Raw on admitted flows):** Retains the exact VLEG admission mask $\mathcal{I}_{\text{adm}}$ and retrieval rankings $\mathcal{M}_{top}$ but strips all reflection text r_i^*, isolating sample-filtering effects from rule-text effects.
- **Multi-Instance Counterexample Screening:** Compiles each admitted IF/THEN rule r_i^* into a deterministic predicate over the feature schema and checks whether its antecedent matches opposite-label records across the remaining historical support pool $\mathcal{E} \setminus \{e_i\}$.

E. Final Algorithm and Execution Flow

Algorithm 1 summarizes the complete two-phase VLEG execution pipeline comprising offline memory consolidation (Phase 1) and online hybrid inference (Phase 2).

By decoupling offline verification from online inference, VLEG ensures that iterative critique overhead is incurred only once during memory bank construction.

Algorithm 1 Verified Localized Episodic Grounding (VLEG)

Input: Episodes $\mathcal{E}$, target T, max attempts K
Retrieval count k, retriever R
Initialize Memory Bank $\mathcal{M} \leftarrow \emptyset$
// Phase 1: Memory Consolidation
for each $e_i \in \mathcal{E}$ **do**
  $j \leftarrow 0$
  $(r, c) \leftarrow (\emptyset, \emptyset)$
  ▷ *Previous candidate and critique*
  $v \leftarrow 0$
  ▷ *Initialize validity flag to Invalid*
  while $v = 0$ **and** $j < K$ **do**
    $r_i^{(j)} \leftarrow G_\theta(x_i, r, c)$
    ▷ *Generate or refine reflection*
    $(v, c) \leftarrow V_\phi(e_i, r_i^{(j)})$
    ▷ *Check reflection against labeled source*
    $r \leftarrow r_i^{(j)}$; $j \leftarrow j + 1$
  end while
  if $v = 1$ **then**
    $\mathcal{M} \leftarrow \mathcal{M} \cup \{(e_i, r_i^{(j-1)})\}$
  ▷ *Store verified tuple*
  end if
end for
// Phase 2: Hybrid Inference
$\mathcal{M}_{top} \leftarrow R(\mathcal{M}, T, k)$
▷ *Retrieve top-k relevant tuples*
Context $\leftarrow$ PromptBuilder$(\mathcal{M}_{top}, T)$
▷ *Construct hybrid prompt*
Output $\leftarrow I_\psi$(Context)
▷ *Generate final reasoning answer*
return Output

IV. Experiments

A. Experimental Settings

We first evaluate VLEG on controlled symbolic Experience-Transfer tasks before moving to the public UNSW-NB15 network flow benchmark. The symbolic testbed defines 16 scene combinations across four causal attributes (shape, color, texture, and background) and a deterministic non-causal distractor (size). On white backgrounds, ANOMALY requires a red circle, whereas black backgrounds invert this parity while leaving texture and size causally irrelevant. Level 1 evaluates eight seen white-background combinations, Level 2 evaluates all 16 seen combinations across both backgrounds, and Level 3 withholds four combinations from memory construction across multiple masking seeds to evaluate out-of-distribution compositional transfer.

Symbolic evaluations use Qwen2.5-7B-Instruct (7B) and Llama-3.3-70B-Instruct (70B) with nomic-embed-text dense embeddings and decoding temperature 0. Initial rule synthesis withholds historical labels, whereas the verifier accesses labeled source episodes and returns natural-language critique across $K \in \{1, 2, 3, 5\}$. All item-level symbolic predictions are verified against archived evaluation snapshots, while our public network flow benchmark further extends the logging pipeline with complete cryptographic hashes, raw verifier responses, and fail-closed native-Boolean JSON parsing.

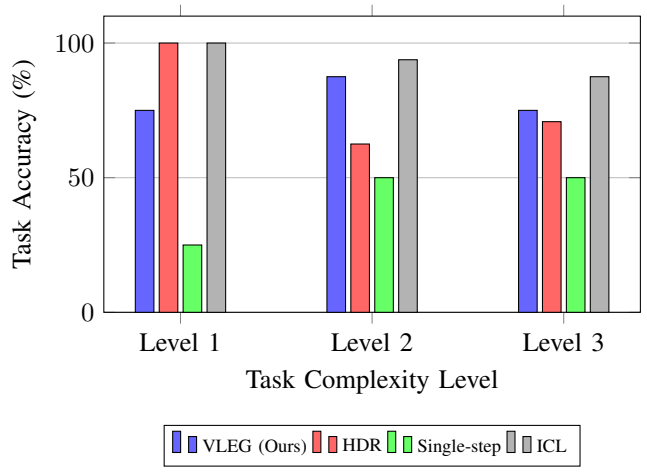


Fig. 4. Measured task accuracy (%) across three compositional complexity tiers on the symbolic benchmark using Qwen2.5-7B-Instruct, with Level 3 pooled across three masking seeds ($n = 48$). Raw ICL leads across all tiers, global hypothesis consolidation (HDR) succeeds on the uniform-background rule in Level 1 but degrades on background-conditional rules in Levels 2 and 3, and unverified single-step reflection tracks the anomaly base rate.

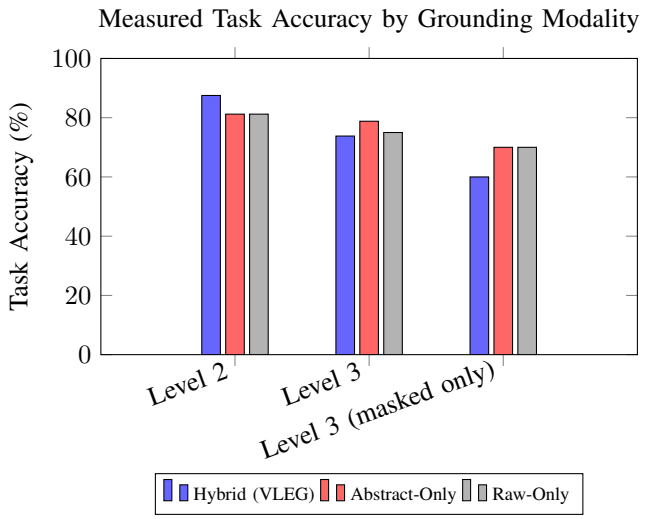


Fig. 5. Measured task accuracy across grounding modalities on Qwen2.5-7B-Instruct, with Level 3 pooled across five masking seeds ($n = 80$ overall and $n = 20$ on the withheld subset). Hybrid grounding achieves the highest accuracy on seen combinations in Level 2 (87.5%) but trails single-modality prompts on the withheld Level 3 subset (60.0% versus 70.0%) due to dual-modality attention competition at the 7B scale.

B. Controlled Memory-Strategy Comparison

Backbone capacity governs whether hybrid verified memory outperforms raw in-context retrieval across compositional difficulty levels. We compare VLEG ($K = 3$) against raw in-context learning (ICL), global hypothesis consolidation (HDR), unverified single-step reflection, and binary-feedback Reflexion across Levels 1–3, where Level 3 pools 48 evaluations across three masking seeds. As shown in Figure 4, on the 7B backbone, Raw ICL leads across all tiers and reaches 87.5% on Level 3 compared with 75.0% for VLEG and 70.8% for global hypothesis consolidation, while historical unverified single-step reflection collapses to the 50.0% anomaly base rate on Levels 2 and 3 because initial label-blind proposals predominantly default to ANOMALY.

In contrast, scaling to the 70B backbone reverses the Level 3 ranking in the reconciled three-seed archive, where VLEG achieves 91.67% accuracy (44/48), edging out Raw ICL at 89.58% (43/48), global hypothesis consolidation at 83.33% (40/48), and binary-feedback Reflexion at 81.25% (39/48). This scale-dependent shift indicates that while smaller backbones struggle to reconcile noisy local abstractions with raw exemplars, larger models can leverage verified local guidance when generalizing across held-out attribute combinations.

C. Component Isolation (Grounding Modality Ablation)

Dual-modality attention competition degrades hybrid grounding on the 7B backbone, whereas the 70B backbone reconciles raw and abstract context effectively. To isolate context formatting from neighbor selection, we fix the VLEG-admitted memory bank and reflection-embedding retrieval indices across five Level 3 masking seeds ($n = 80$ total evaluations, $n = 20$ on withheld combinations) and compare Hybrid (VLEG), Abstract-Only, and Raw-Only prompts. As illustrated in Figure 5, on the 7B backbone Hybrid grounding achieves the highest accuracy on seen combinations in Level 2 (87.5%) but drops to 60.0% on the withheld Level 3 subset, trailing both Abstract-Only and Raw-Only by 10.0 pp.

When evaluated on the 70B backbone across the same five masks, the 7B overall deficit disappears as Hybrid and Raw-Only tie at 95.0% overall accuracy ahead of Abstract-Only at 91.25%, while Abstract-Only retains a 5.0 pp edge on the 20 withheld instances (85.0% versus 80.0%). This contrast occurs because the 7B model suffers from dual-modality attention competition when forced to process raw attribute tuples alongside imperfect rules that still cite non-causal distractors, whereas the 70B backbone possesses sufficient contextual capacity to integrate both representations without catastrophic interference.

D. Iterative Refinement Ablation

Iterative verification improves downstream accuracy primarily by recovering memory-bank coverage rather than purifying causal attributions. Comparing strict verification at $K = 1$ (where any rejected initial proposal discards the episode) against $K = 3$ isolates the effect of iterative critique. Across five 7B masking seeds, extending the budget from $K = 1$ to $K = 3$ lifts overall Level 3 accuracy by 11.25 pp from 63.75% (51/80) to 75.0% (60/80) and raises withheld-combination accuracy by 10.0 pp from 50.0% to 60.0%, while on the 70B backbone $K = 3$ yields a mean accuracy gain of 26.2 ± 8.1 pp across the five masks.

The primary physical driver of this gain is memory-bank coverage recovery. At $K = 1$, initial label inversions cause the verifier to reject 42% to 58% of historical support episodes on the 7B model, starving the retriever of critical reference combinations. Allowing up to $K = 3$ refinement attempts enables verifier critique to repair mismatched historical labels, restoring 7B memory admission coverage to between 83%

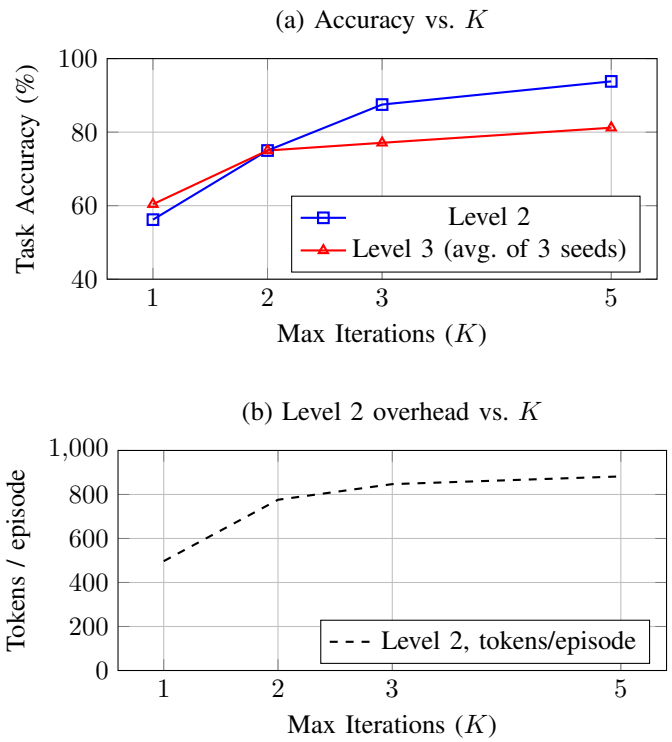


Fig. 6. Parameter sensitivity of the maximum refinement budget K on Qwen2.5-7B-Instruct. (a) Task accuracy across refinement budgets for Level 2 and Level 3 (averaged over three masking seeds), showing progressive coverage recovery through $K = 5$ at the 7B scale. (b) Consolidation token overhead per historical episode on Level 2, exhibiting concave growth in which the primary marginal cost occurs between $K = 1$ and $K = 2$ because most episodes resolve within two verification attempts.

and 100% and achieving 100% admission across all 12 support episodes per seed on the 70B model. Because iterative critique repairs source-label agreement without introducing cross-episode contrast, recovered reflections expand retrieval coverage while remaining subject to single-instance causal blindness.

E. Parameter Sensitivity and Overhead Analysis

Refinement token overhead grows concavely with K because most candidate reflections resolve within two verification attempts. To evaluate the trade-off between classification accuracy and offline bank construction cost, we sweep the refinement ceiling across $K \in \{1, 2, 3, 5\}$ and record both task accuracy and total prompt-plus-completion tokens per historical episode.

As shown in Figure 6, token overhead per consolidated episode increases concavely rather than linearly on the 7B backbone, adding 279 tokens per episode from $K = 1$ to $K = 2$, 71 tokens from $K = 2$ to $K = 3$, and only 35 tokens from $K = 3$ to $K = 5$. Because the loop terminates immediately upon first acceptance, raising K expends additional compute only on the minority of episodes that fail early checks. While 7B accuracy continues to edge upward through $K = 5$ as residual rejected episodes are recovered, the 70B backbone reaches an exact plateau at $K = 2$ across all three Level 3 seeds, admitting 100% of support episodes with identical token counts and accuracy at $K = 2$, 3, and 5. Consequently, a modest refinement budget of $K = 2$ or 3 captures the vast majority of coverage gains at bounded offline cost.

TABLE I
DETERMINISTIC MEMORY-QUALITY DIAGNOSTIC ON QWEN2.5-7B-INSTRUCT. ADMISSION DENOTES THE PROPORTION OF HISTORICAL SUPPORT EPISODES ACCEPTED WITHIN $K = 3$ ATTEMPTS, LABEL-CORRECT DENOTES SOURCE-LABEL AGREEMENT AGAINST THE GROUND-TRUTH ORACLE, AND SPURIOUS (OF CORRECT) MEASURES THE PERCENTAGE OF LABEL-CORRECT REFLECTIONS CITING THE NON-CAUSAL ATTRIBUTES TEXTURE OR SIZE.

Level	Strategy	Admission	Label-correct	Spurious (of correct)
Level 2	Single-Step	N/A	50.0% (8/16)	87.5% (7/8)
	VLEG (K=3)	**93.8%**	**100.0% (15/15)**	86.7% (13/15)
Level 3	Single-Step	N/A	41.7% (5/12)	80.0% (4/5)
	VLEG (K=3)	**100.0%**	**100.0% (12/12)**	83.3% (10/12)

F. Diagnostic Assessment of Memory Quality

Single-instance verification enforces 100% source-label consistency but cannot filter co-occurring non-causal distractors. To directly test what localized verification filters during memory construction, we deterministically audit admitted reflections along two orthogonal axes, namely source-label agreement against the ground-truth episode label and spurious attribution rates measuring whether label-correct rules cite the non-causal distractors texture or size.

Table I demonstrates that on the 7B backbone, VLEG ($K = 3$) raises source-label consistency from at most 50.0% under unverified single-step generation to 100.0% across all admitted memories at both Level 2 and Level 3. However, conditional on label correctness, VLEG and unverified single-step generation cite non-causal distractors at nearly identical rates, reaching 86.7% versus 87.5% on Level 2 and 83.3% versus 80.0% on Level 3.

This diagnostic confirms the structural boundary formalized in Equation (5). Because the verifier inspects only the single originating episode e_i, any incidental attribute that physically co-occurs on e_i (such as a striped texture on an anomalous red circle) is factually true for that instance and therefore passes single-record verification. On the 70B backbone, source-label consistency similarly rises from 50.0% for unverified first proposals to 100.0% for VLEG, while unconditional distractor mention rates drop to 12.5% (2/16) on Level 2 and 16.7% (2/12) on Level 3 across both verified and unverified strategies alike. Thus, lower distractor citation at 70B stems from stronger base-model priors rather than single-instance verification, demonstrating why cross-episode contrast is required to eliminate spurious covariates.

G. Public-Flow Benchmark Validation

Experimental Protocol and Leakage-Free Preprocessing. We next transfer our auditing pipeline to the public UNSW-NB15 network intrusion testbed [20] after applying a 19-feature deterministic whitelist, removing exact duplicate feature vectors, excluding conflicting-label records, and enforcing

zero overlap between historical support pools and test queries. Across five historical memory seeds, each run samples 32 balanced support flows (16 benign and 16 malicious) and evaluates a shared set of 1,000 disjoint test flow fingerprints (663 benign and 337 malicious, yielding 25,000 audited formal predictions across five conditions) on a dual-A800 server running Llama-3.3-70B (Q4_K_M) and nomic-embed-text at temperature 0 with $K = k = 3$. To isolate memory consolidation from neighbor selection, Raw flow retrieval, Unverified single-step, VLEG, and our control condition Raw on admitted flows share identical label-free feature retrieval rankings, where rejected episodes are omitted without replacement and Raw on admitted flows retains VLEG's exact admission mask while stripping all reflection text.

VLEG prevents the catastrophic attack-recall collapse of unverified reflection by correcting benign bias during memory admission. As shown in Table II and Figure 7, Unverified single-step reflection suffers a severe collapse in attack recall to 6.17%, yielding a Macro-F1 of only 42.66%. Trajectory inspection reveals that when the label-blind generator encounters common connection states such as TCP FIN termination, it defaults to benign predictions and stores normal-labeled rules for actual attack episodes. By enforcing label-informed verification over up to $K = 3$ steps, VLEG admits 98.75% of candidate memories ($158/160$ across the five seeds), repairs the inverted attack labels, lifts attack recall by 71.57 pp to 77.74%, and improves Macro-F1 by $+17.02$ pp within a paired conditional 95% bootstrap confidence interval of $[13.71, 20.39]$ pp.

Locally valid IF/THEN rules over-generalize single-flow states into globally sufficient triggers, inflating false positives relative to raw flow retrieval. Despite outperforming unverified reflection and Global summary (47.60% Macro-F1 and 68.54% false positive rate), VLEG trails pure Raw flow retrieval (63.92% Macro-F1) by 4.24 pp within a paired conditional 95% confidence interval of $[-5.91, -2.53]$ pp. Comparing VLEG against Raw on admitted flows (63.70% Macro-F1, within 0.22 pp of full raw retrieval) proves that rejecting 2 of 160 support episodes is not the cause of this gap. Rather, our deterministic rule-screening compiler reveals that while 157 of the 158 admitted VLEG rules parse cleanly and achieve 100% factual and label agreement on their source flow, 113 of those 157 locally valid rules (72.0%) match opposite-label counterexamples among the remaining historical flows. Appending these overly broad IF/THEN heuristics increases attack recall by 6.35 pp over raw retrieval at the cost of raising the false positive rate by 10.65 pp from 38.13% to 48.78%. Table III summarizes the corresponding memory construction and query latency costs across all five conditions.

H. Limitations

Our empirical findings are subject to explicit domain and provenance boundaries. First, the symbolic testbed evaluates a compact 16-combination population where the ground-truth rule is expressible as a single background-conditioned parity statement, and cross-scale comparisons between Qwen2.5-7B and Llama-3.3-70B change model families alongside parameter count. Second, our UNSW-NB15 evaluation operates on

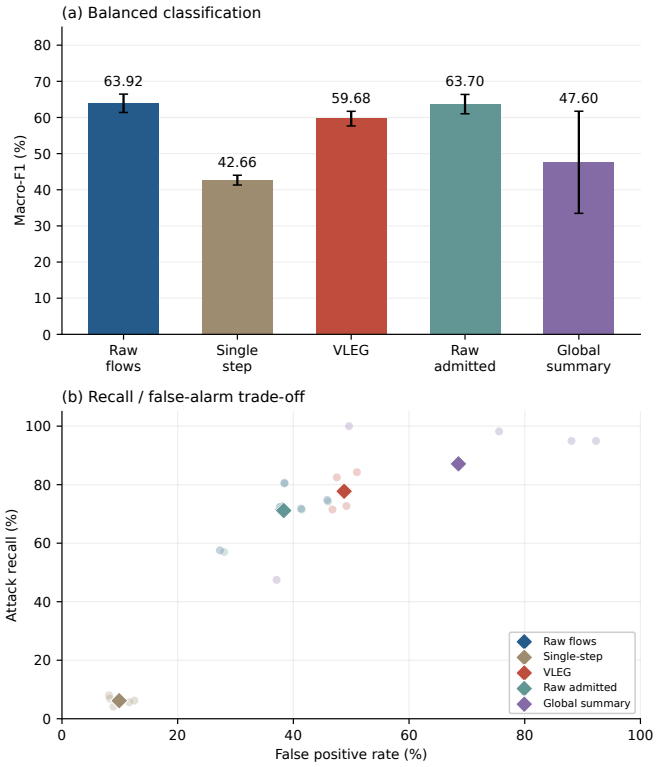


Fig. 7. UNSW-NB15 evaluation across five historical-memory seeds, showing mean Macro-F1 with sample-standard-deviation error bars (top) and attack recall versus false positive rate across individual seeds and condition means (bottom).

a deduplicated, conflict-free 19-feature partition under shared label-free feature retrieval, fixed method execution order, and five support seeds over a single 1,000-flow test split.

Furthermore, whereas the symbolic testbed relies on standard JSON verification parsing over synthetic attributes, our UNSW-NB15 pipeline introduces fail-closed native-Boolean validation alongside an automated rule compiler that deterministically screens 99.4% ($157/158$) of admitted traffic reflections against historical counterexamples. Extending this automated counterexample screen with domain-expert human blind review across multi-campus traffic captures remains an important direction for future operational validation.

TABLE II
PUBLIC NETWORK FLOW CLASSIFICATION PERFORMANCE ON UNSW-NB15 (MEAN ± SAMPLE STANDARD DEVIATION IN PERCENT ACROSS FIVE HISTORICAL-MEMORY SEEDS ON 1,000 SHARED TEST FINGERPRINTS). UNPARSEABLE OUTPUTS ARE SCORED AS FALSE NEGATIVES FOR THEIR GROUND-TRUTH CLASS IN MACRO-F1 AND RECALL, WHILE THE FALSE POSITIVE RATE MEASURES BENIGN FLOWS PREDICTED AS MALICIOUS.

Condition	Accuracy	Macro-F1	Attack recall	False positive rate
Raw flow retrieval	65.08 ± 2.93	63.92 ± 2.54	71.39 ± 8.44	38.13 ± 6.85
Unverified single-step	61.38 ± 1.08	42.66 ± 1.35	6.17 ± 1.43	9.89 ± 2.03
VLEG	59.74 ± 2.03	59.68 ± 2.03	77.74 ± 5.69	48.78 ± 1.67
Raw on admitted flows	64.86 ± 2.96	63.70 ± 2.68	71.16 ± 8.71	38.34 ± 6.61
Global summary	50.22 ± 12.45	47.60 ± 14.13	87.12 ± 22.27	68.54 ± 24.20

TABLE III
OBSERVED PUBLIC-FLOW COSTS, MEANS ACROSS FIVE MEMORY SEEDS. CONSTRUCTION TOKENS INCLUDE INPUT AND OUTPUT. CHAT LATENCY EXCLUDES SHARED RETRIEVAL AND IS CONFOUNDED BY FIXED METHOD ORDER/CACHE REUSE.

Condition	Construction calls	Construction tokens	Construction (s)	Query mean (s)	Query p95 (s)
Raw flow retrieval	0	0	0	2.12	2.67
Unverified single-step	32.0	9,405	126.54	2.23	2.75
VLEG	120.4	40,094	419.43	1.74	2.68
Raw on admitted flows	120.4	40,094	419.43	0.95	1.19
Global summary	1.0	5,947	23.21	1.21	1.30

V. CONCLUSIONS

This paper presented Verified Localized Episodic Grounding (VLEG) and conducted a controlled mechanistic study spanning symbolic compositional reasoning and public UNSW-NB15 network traffic classification. Our results establish that interposing a bounded, label-informed verification loop prior to memory admission eliminates source-label inversions, guarantees 100% source-episode label consistency, and restores memory-bank coverage within two to three refinement attempts. On the 70B backbone, these properties enable VLEG to achieve 91.67% accuracy on held-out symbolic combinations and improve UNSW-NB15 Macro-F1 by 17.02 pp over unverified single-step reflection by recovering attack recall from 6.17% to 77.74%. Simultaneously, our admitted-mask control and deterministic counterexample screen prove that local source agreement does not guarantee cross-episode specificity, because 72.0% of single-flow verified IF/THEN rules fire on opposite-label historical flows and elevate false alarms relative to raw flow retrieval.

For network analysts and systems practitioners, these findings yield three actionable deployment principles. First, whenever natural-language reflection is required for human auditability, it must be paired with label-informed verification ($K = 2$ or 3) to prevent severe benign bias and recall collapse. Second, when operating on low-dimensional structured flow features without a multi-instance verifier, direct raw exemplar retrieval remains preferable because it avoids sufficiency overgeneralization at zero offline LLM construction cost. Third, future reflective network agents should embed deterministic multi-instance counterexample screening directly inside the verification gate to reject or tighten candidate rules that match opposite-label historical traffic before committing them to long-term memory.

REFERENCES

[1] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, S. Agarwal, A. Herbert-Voss, G. Krueger, T. J. Henighan, R. Child, A. Ramesh, D. M. Ziegler, J. Wu, C. Winter, C. Hesse, M. Chen, E. Sigler, M. Litwin, S. Gray, B. Chess, J. Clark, C. Berner, S. McCandlish, A. Radford, I. Sutskever, and D. Amodei, "Language Models are Few-Shot Learners," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 1877–1901.

[2] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Advances in Neural Information Processing Systems*, H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, and H. Lin, Eds., vol. 33. Curran Associates, Inc., 2020, pp. 9459–9474.

[3] C. An, J. Zhang, M. Zhong, L. Li, S. Gong, Y. Luo, J. Xu, and L. Kong, "Why Does the Effective Context Length of LLMs Fall Short?" *ArXiv*, vol. abs/2410.18745, 2024.

[4] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang, "Lost in the Middle: How Language Models Use Long Contexts," *Transactions of the Association for Computational Linguistics*, vol. 12, pp. 157–173, 2024.

[5] C. Packer, S. Wooders, K. Lin, V. Fang, S. G. Patil, I. Stoica, and J. E. Gonzalez, "MemGPT: Towards LLMs as Operating Systems," *ArXiv*, vol. abs/2310.08560, 2023.

[6] W. Wang, L. Dong, H. Cheng, X. Liu, X. Yan, J. Gao, and F. Wei, "Augmenting Language Models with Long-Term Memory," in *Advances in Neural Information Processing Systems*, vol. 36, 2023.

[7] Y. Wang, Y. Gao, X. Chen, H. Jiang, S. Li, J. Yang, Q. Yin, Z. Li, X. Li, B. Yin, J. Shang, and J. McAuley, "MEMORYLLM: Towards Self-Updatable Large Language Models," in *Proceedings of the 41st International Conference on Machine Learning*, vol. 235, 2024, pp. 50 453–50 466.

[8] Y. Wu, S. Liang, C. Zhang, Y. Wang, Y. Zhang, H. Guo, R. Tang, and Y. Liu, "From Human Memory to AI Memory: A Survey on Memory Mechanisms in the Era of LLMs," *ArXiv*, vol. abs/2504.15965, 2025.

[9] Z. Zhang, X. Bo, C. Ma, R. Li, X. Chen, Q. Dai, J. Zhu, Z. Dong, and J.-R. Wen, "A Survey on the Memory Mechanism of Large Language Model based Agents," *ArXiv*, vol. abs/2404.13501, 2024.

[10] N. Shinn, F. Cassano, A. Gopinath, K. R. Narasimhan, and S. Yao, "Reflexion: language agents with verbal reinforcement learning," in *Thirty-seventh Conference on Neural Information Processing Systems*, 2023.

[11] G. Wang, Y. Xie, Y. Jiang, A. Mandlekar, C. Xiao, Y. Zhu, L. Fan, and A. Anandkumar, "Voyager: An Open-Ended Embodied Agent with Large Language Models," *Transactions on Machine Learning Research*, 2024.

[12] A. K. Agarwal and M. Yamada, "Hypothesis-Driven Reasoning for Large Language Models," in *Proceedings of the Fortieth AAAI Conference on Artificial Intelligence (AAAI-26)*, 2026, pp. 1686–1693.
[13] K. He, M. Zhang, S. Yan, P. Wu, and Z. Chen, "IDEA: Enhancing the Rule Learning Ability of Large Language Model Agent through Induction, Deduction, and Abduction," in *Annual Meeting of the Association for Computational Linguistics*, 2024.
[14] Y. Zhang, S. Feng, and C. Tan, "Active Example Selection for In-Context Learning," *ArXiv*, vol. abs/2211.04486, 2022.
[15] K. Purohit, V. V, S. Bhattacharya, and A. Anand, "Sample Efficient Demonstration Selection for In-Context Learning," in *Forty-second International Conference on Machine Learning*, 2025.
[16] W. Zhong, L. Guo, Q. Gao, H. Ye, and Y. Wang, "MemoryBank: Enhancing Large Language Models with Long-Term Memory," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 17, pp. 19 724–19 731, 2024.
[17] J. S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative Agents: Interactive Simulacra of Human Behavior," in *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, 2023.
[18] W. Xu, Z. Liang, K. Mei, H. Gao, J. Tan, and Y. Zhang, "A-MEM: Agentic Memory for LLM Agents," *ArXiv*, vol. abs/2502.12110, 2025.
[19] S. H. Kim, K. T. Ong, T. Kwon, N. Kim, K. Ka, S. Bae, Y. Jo, S. Hwang, D. Lee, and J. Yeo, "Towards Lifelong Dialogue Agents via Timeline-based Memory Management," in *North American Chapter of the Association for Computational Linguistics*, 2024.
[20] N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set)," in *Military Communications and Information Systems Conference (MilCIS)*, 2015.